

Joshua Negreanu

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PhD student researching reinforcement learning for agentic language models, with hands-on experience building model, training, and data infrastructures at scale.

EDUCATION

- Oregon State University** Corvallis, Oregon
MS Artificial Intelligence (*Accelerated*), GPA: 4.00 Sep 2025 - Jun 2027
- Oregon State University** Corvallis, Oregon
BS Computer Science (*AI focus*), GPA: 4.00 Sep 2021 - Jun 2025
BS Mathematics (*Applied focus*), GPA: 4.00 Sep 2021 - Jun 2025

SKILLS SUMMARY

- Languages:** Python, C/C++, JavaScript, Bash, SQL, Java
- ML/DL:** PyTorch, Hugging Face (LeRobot, Transformers), NumPy, SciPy, Scikit-learn, Wandb
- Systems:** CUDA, OpenMP, OpenCL, GStreamer, Linux, Git, Docker
- Data/Infrastructure:** Pandas, Matplotlib, OpenCV, AWS, Azure, DGX, Jetson
- Backend:** Node.js, mediasoup, vLLM, Modal, WebXR

EXPERIENCE

- Revobots** Remote
AI Intern Jun 2025 - Feb 2026
 - Remote Teleoperation:** Designed and deployed mediasoup selective forwarding unit server for low-latency (30 fps) robot vision and control streaming, supporting 500+ concurrent clients per robot.
 - VR Embodiment:** Implemented 60-fps 4K stereo streaming with WebXR headset control and GStreamer GPU encoding for real-time VR embodiment and dataset collection.
 - Data Collection:** Built end-to-end training data pipeline using Hugging Face LeRobot for teleoperation recording, collecting 100+ episode datasets with automated compression, AWS storage, and metadata tagging.
 - Model Training:** Designed MLOps pipeline for automated multi-day training of transformer-based manipulation policies on DGX, learning from collected demonstration trajectories via imitation learning, achieving reliable task execution.
- Oregon State University** Onsite
Educational Assistant Sep 2022 - Feb 2026
 - Teaching:** Graduate and undergraduate teaching assistant across four courses (Programming Language Fundamentals, Operating Systems, Data Structures, General Engineering), served 100+ students total with office hours, grading hours, and recitations. Wrote autograder scripts for unit testing functional programming assignments in Haskell and Prolog, reducing manual grading time by 70%.

RESEARCH

- Undergraduate Thesis: Attacks on Stochastic Dueling Bandits:** Proposed a mathematical framework for reward poisoning algorithms in stochastic dueling bandits setting, demonstrating vulnerabilities that motivated development of corruption-robust algorithms. Under the mentorship of Huazheng Wang. Paper available on *OSU Archives*.

PROJECTS

- Masked Diffusion and Autoregressive Language Models:** Built both traditional autoregressive and masked diffusion transformer language models from scratch. Scaled to 350M parameters on cosmopedia-v2, supervised finetuning of an instruct variant on smoltalk, deployed on a chat webapp with scale-to-zero model hosting: <https://joshnegreanu.com/chat>. Open-source code available on Github: <https://github.com/joshgpt-llm>.
- Thompson Sampling with Diffusion and Flow Matching Priors for Linear Bandits:** Developed both guided diffusion and flow matching algorithms for linear bandit prior sampling for applications in Thompson sampling algorithms, as well as low-rank approximations to reduce per-round time complexity by 75% while preserving regret performance within 5%. Open-source code available on GitHub: <https://github.com/joshnegreanu/diffusion-priors>.
- AI Image Detection App (Hackathon Project):** Worked in team of 4 to finetune NVIDIA Nemotron 12B vision-language model for AI image detection, served on Lambda with vLLM, packaged as an Electron app with screenshot-detection capabilities. Open-source code available on Github: <https://github.com/bobtheblb/beav-hacks-2026>.
- Denoising Diffusion and Flow Matching Image Generation:** Implemented denoising diffusion and flow matching generative models (CNN U-Net and latent vision transformer) from scratch. Trained on Stanford Cars, CelebA, and ImageNet on DGX system. Open-source code available on GitHub: <https://github.com/joshnegreanu/image-diffusion>.

HONORS AND AWARDS

- CV Copson Scholarship for Mathematics:** Awarded at annual OSU Lonseth Lecture, May 2025
- WIC Culture of Writing Award for Mathematics:** Awarded at annual OSU Lonseth Lecture, May 2025
- Dean's List:** Scholastic Honor Roll (3.5+ GPA), Sep 2021 - Jun 2025